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Patient Flow Optimization in an Emergency Department Using SimPy-Based Simulation Modeling and Analysis: A Case Study

Anudeep Battu, S. Venkataramanaiah, and R. Sridharan

Abstract Long waiting times and patient congestion are common problems faced by emergency departments (EDs) worldwide. During pandemics like COVID-19, EDs worldwide start to be flooded with patients and hospitals find it very challenging to provide good treatment to the large number of patients visiting the EDs with their current allocation of resources. Hospitals are in need of a decision support system (DSS) which can predict the excess demand and suggest the appropriate quantity of resources to be allocated at each point of care. The present research focuses on an ED of a large public hospital in India and explores in finding a solution for the long patient waiting time problem experienced by the hospital. This study extends the application domain of SimPy-based simulation modeling with integrated metamodeling and optimization to optimally allocate the resources in the ED. This can be used as a novel DSS which is relatively faster and needs less human interaction by the hospital management compared to the existing methods. The proposed resource allocation by this model reduced the patient waiting time by 44% in the case hospital being studied. Hospitals may use the proposed methodology to appropriately allocate their resources in times of excess demand.

Keywords Emergency department · Bottlenecks · Discrete event system simulation · Multiple linear regression · Optimization · SimPy

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1 Introduction

Reduced waiting time of patients and management of resources in the emergency departments (EDs) is very crucial for any hospital. In recent years, people started giving more priority to health care than anything else. During the recent COVID-19 pandemic, many countries have failed in managing their COVID-19 cases and the hospitals started flooding with emergency patients who need critical treatment and resources like ventilators and oxygen support. The emergency departments never experienced such a situation before. Hence, the patients had to wait for a long time and many patients have died due to lack of proper treatment and resources in hospitals. Long waiting times and patient's congestion are few of the biggest problems faced by hospitals worldwide. In the recent times, the number of patients visiting the EDs has also increased drastically. Thus, there is a need for finding a solution to this problem.

According to the literature review, the most appropriate methodology for determining optimal resource allocation and reducing patient waiting time in EDs is simulation modeling. Simulation modeling helps in assessing the consequences of making changes in the system without disturbing the system.

The remaining portion of the paper is organized in the following way. In the next section, salient and relevant literature belonging to the present research domain is reviewed briefly. In Sect. 3, the research methodology is presented. The details of the simulation model are presented in Sect. 4. Metamodel-based optimization is described in detail in Sect. 5. Results and interpretation are provided in the next section. Finally, in the last section, conclusions and scope for future work are articulated.

2 Literature Review

Several researchers have adopted simulation modeling for investigating the healthcare services. Khekale et al. [1] provided an extensive review on the use of simulation models in reducing the patient waiting times in the EDs. The researchers describe the salient aspects of the analysis of various studies based on their modeling techniques, data sources, collection methods and study findings. A comprehensive review of published studies from each category along with a discussion on their important features is furnished.

Ahmed et al. [2] presented a study in which a decision support tool is designed by integrating simulation with optimization. The optimum number of resources required such as the nurses, laboratory technicians, doctors, etc., can be found out using the simulation-based optimization methodology proposed in this study. Using the proposed method, optimization can be done to improve the patient throughput and to reduce the time patient spends in the system considering the budget constraints.

Cabreraa et al. [3] adopted simulation and agent-based modeling to design a decision support system which optimizes the processes, resources and operations of

emergency departments and aids the hospital management in formulating strategies for optimization.

Lin et al. [4] conducted a study in which the power of genetic algorithm (GA) and data envelopment analysis (DEA) are used for proposing a multiobjective simulation and optimization framework. This method can be used for searching very large design spaces effectively. In this method, GA is selected using the fitness value as the value obtained from the relative efficiency score of design point from data envelopment analysis. Then, the algorithm is applied for optimization of resources in surgical services.

Zeinali et al. [5] carried out a work which portrays a novel tool which can be used for decision support in all the three levels (i.e.,) strategic, tactical and operational. The model proposed by the authors can be used to reduce the total average waiting time of ED patients taking budget and capacity constraints into consideration.

Thim et al. [6] investigated the significant factors affecting the length of stay of patient in the hospital, and the findings suggest the use of lean principles for process improvement in the EDs in hospitals. They propose setting up referral processes, using third-party coordinators and setting up 24-h ambulance services for integrating the processes from the site at which the accident occurred to the emergency department in the hospital.

From the literature review, the following inferences can be drawn:

- Although several studies have been reported on simulation of emergency departments, there are only a few studies on trauma care in Indian hospitals. Indian healthcare system is different from others because low-cost trauma care is to be provided to geographically large and populated areas.
- Reducing patient waiting time is the primary objective in most studies performed.
- Data collected in most studies for simulation modeling involves patient interarrival times, healthcare service times, length of stay, volume and mix of patients. Data collection in most studies uses time and motion studies.
- The findings from most studies can be categorized in two ways: resource related and process related. Optimization of resource allocation has been pursued in relatively more studies than optimization of process.

3 Research Methodology

The objective of the present study is to develop a novel decision support system to reduce the patient waiting time in emergency departments of hospitals. In this study, firstly data regarding various parameters to be input in the simulation model such as patient arrival rate, service times at various points of care and transit times is obtained from a previous study performed in the hospital. In the next step, a simulation model is developed using SimPy library in Python programming language. The results are compared with that obtained from a simulation model developed using the process-oriented simulation package, Arena. In the next step, the simulation model is run for different configuration of resources and the average patient waiting time is obtained

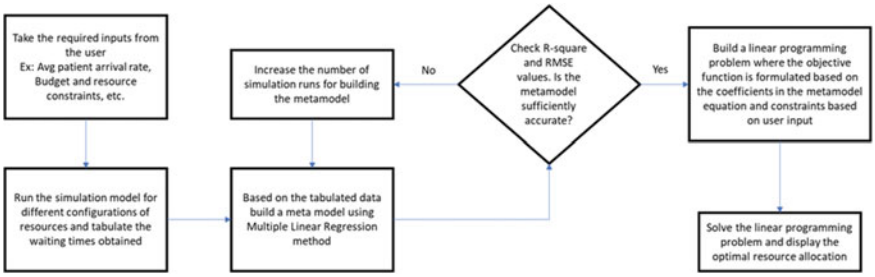

Fig. 1 Simulation-based metamodeling and optimization using SimPy

for each configuration. This data is tabulated into a data frame in Python. Based on the tabulated data, a multiple linear regression (MLR) metamodel is developed and the accuracy of the model is checked. Using the coefficient values obtained from MLR, a metamodel equation is obtained. In the next step, a linear optimization problem is formulated based on the metamodel equation and resource and budget constraints specified by the hospital management with the objective of reducing the average patient waiting time. The quantity of resources to be allocated at various points of care is the decision variables. The present study focuses on obtaining the optimal resource allocation at all points of care in the ED using this research methodology (Fig. 1).

4 Simulation Model Development

The emergency department of a large public hospital located in the northern part of India is considered in the present study. When a patient enters the emergency department, a stretcher is located and brought to the entrance of the emergency department. The patient is then kept over the stretcher and moved to the emergency room. In the emergency room, an ED doctor and paramedical staff examine the patient for movements and the patient is later sent for laboratory investigations (X-ray, MRI and ultrasound). The attendants accompanying the patient are issued the bill for laboratory investigations in the cash counter. After the laboratory investigation, the patients are re-examined by an ED doctor and paramedical staff. The patients are then sent to the surgery ward for opinion. The surgeon inspects and sends the patient to the specialty ward. Specialists check the patient and send him to the operation theater. The orthodoctor performs the surgery, and a bed is allocated to the patient in the in-patient department. After observation and treatment for a few days, the patient is discharged. In this study, the objective is to reduce the patient waiting time from the entry stage to the surgery stage. The process chart of processes followed by various types of patients from entry to surgery is shown in Fig. 2.

The main resources and their quantities used in the emergency department are as follows:

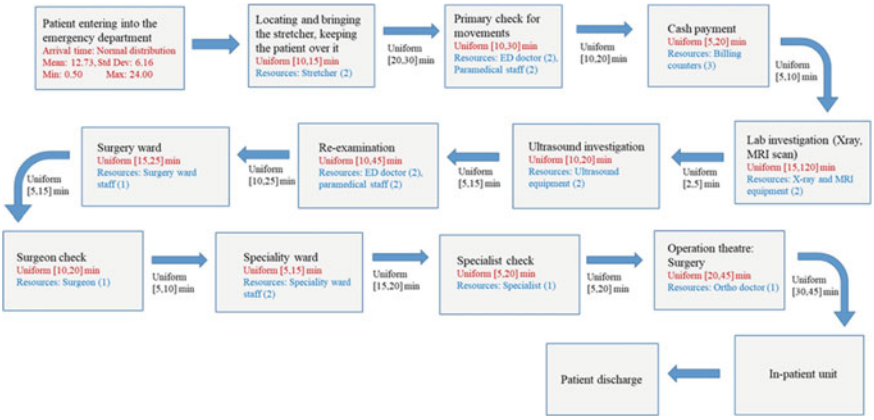

Fig. 2 Process flowchart of the emergency department

- Stretcher (2)
- ED doctor (2) and paramedical staff (2)
- Billing counters (3)
- X-ray, ultrasound and MRI equipment (2)
- Surgeon (1) and surgery ward staff (1)
- Specialist (1) and specialty ward staff (2)
- Orthodoctor (1) and operation theater staff (1)

Two simulation models of the emergency department were developed, one using Arena 15.0 simulation package and the other using SimPy library in Python. The key inputs required for the simulation model are the following: (1) interarrival time distribution, (2) service time distribution at various points of care, (3) transit time distribution between various points of care and (4) quantity of resources used at each point of care. Many of the inputs were collected from a previous study done in the emergency department. In the present study, around 104 patients were given a questionnaire survey and data regarding their arrival time service times, transit times and wait times at various points of care was obtained. Distributions regarding the inputs in the simulation model were obtained using the data collected. The interarrival time, the service and the transit time distribution at various points of care are specified in Fig. 2.

The warm-up period was derived as 160 days using the Welch's method [7]. The warm-up period is set as 160 days before collecting the results from the model for the queues to settle into normal running conditions. The run length and number of replications were derived using the desired half-width of 5% of the average length of stay (LOS). Accordingly, the simulation models were run for 360 days with 100 replications.

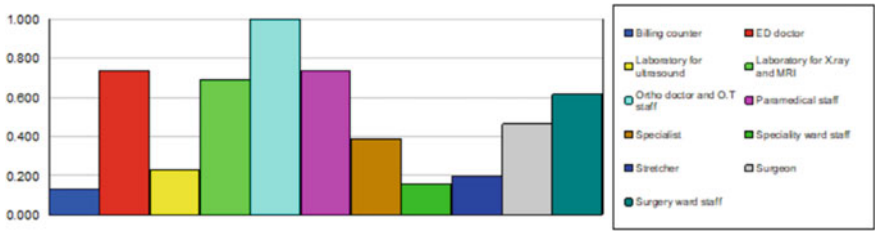

Fig. 3 Utilization rates of various resources

Verification is the process of ensuring that the model is built correctly. In the present study, verification is done by comparing the results obtained from the simulation models to the performance data obtained from the actual emergency department. Moreover, the hospital managers have confirmed that the simulation model is able to accurately represent the emergency department and can be used to assess the changes in the system to make managerial decisions.

Validation is the process of determining whether the right model is built for the right problem. The simulation model is run using different data and different configurations of resources, and the model is found to be sufficiently accurate.

5 Metamodel-Based Optimization

5.1 Multiple Linear Regression Metamodel

Multiple linear regression (MLR) models are widely used to determine the relationship between a single dependent variable and two or more independent variables. By using MLR models, results can be tabulated and the hospital management can draw inferences without much prior knowledge in statistical methods. In this study, linear regression is used to determine the effect of altering the quantity of resources at various stages on the total average length of stay (LOS) of a patient. The aim of linear regression is to construct the best-fitting straight-line to the observed data and using least squares method to determine the regression coefficients which define the line. A multiple linear regression equation with three independent variables and one target variable can be described as follows:

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + e$$

where Y is the target variable, β_0 is the y-intercept, β_1, β_2, β_3 are the regression coefficients, x_1, x_2, x_3 are the independent variables and e is the error term. The data required for building the metamodel was collected from the SimPy simulation model by repeatedly running the model and collecting the results for different sets of resource configurations. The dataset was divided into training data and testing

data using Scikit-learn library by taking 80% data for training and 20% for testing. The multiple linear regression model was fit using Statsmodels library in Python, and results were obtained in a tabular form. From the results, different regression coefficients were obtained and the model is derived. The measures, namely coefficient of determination (R^2) and root mean square error (RMSE) of the multiple linear regression models, were obtained for verifying the accuracy of the model.

5.2 Optimization

Linear optimization is done using PuLP library in Python to obtain the optimal resource allocation to reduce the waiting time of patients in the emergency department subject to various capacity and budget constraints of the department. The mathematical model of the optimization problem is as follows:

$$\begin{aligned}
&\text{Minimize } Z = f(X_1, X_2, X_3, \ldots, X_n)\\
&\text{Subject to}\\
&\sum C_i X_i \leq B\\
&L_i \leq X_i \leq U_i \quad \text{for } i = 1 \text{ to } n\\
&X_i \text{ is an integer for } i = 1 \text{ to } n
\end{aligned}$$

This is an integer programming problem where Z represents the objective function value, i.e., the total average length of stay (LOS) of a patient, X_i represents the quantity of resources at the ith point of care and n represents the total number of points of care in the ED, C_i represents the cost incurred by the hospital in increasing the corresponding resource by 1 unit at the ith point of care, L_i and U_i represent the minimum and maximum quantities of resources feasible by the hospital at the ith point of care and $f(X_i)$ represents the metamodel equation derived using the coefficients obtained from the MLR model.

6 Results and Discussion

It is observed from the simulation results in Table 1 and (Fig. 3) that the average waiting time of patients and utilization rates of resources are very high at various points of care in the ED. The significant bottlenecks identified in the ED are at the laboratory investigation stage, surgery ward stage, specialty ward stage and operation theater stage with waiting time of more than 1 h. The total average waiting time by patients is observed to be around 2.27 days. Table 2 provides the results of the MLR metamodel. From Fig. 4, the condition that the error terms in the MLR model are normally distributed is supported as the normal probability plot is approximately linear. The R-square and adjusted R-square values for the MLR model are obtained

Table 1 Waiting times at various points of care

Point of care (POC)	Average waiting time (minutes)
Stretcher	25
primary check	45
Cash payment	30
Laboratory investigation	180
Ultrasound Investigation	20
Re-examination	30
Surgery ward	110
Surgeon check	75
Specialty ward	185
Specialist check	52.5
Operation theater	2520

as 0.710 and 0.684, respectively, which implies that there are only small differences between the observed and the fitted values. Thus, the model is sufficiently accurate. Table 3 provides the details of the proposed resource allocation obtained from the model. The proposed resource allocation reduces the total average patient waiting time by almost 1 day, i.e., 44% reduction in total waiting time. The results from the SimPy-based metamodeling and optimization model are validated by comparing that obtained from the model developed for the proposed resource configuration using the simulation software Arena. Furthermore, the proposed system is observed to be faster and inexpensive because SimPy is a library in a popular programming language named Python. The Simpy software can be used and distributed freely (even for commercial use) since it is developed under open-source license approved by the open-source initiative.

7 Conclusion

The present study applies simulation, metamodeling and optimization for appropriate resource allocation in hospital EDs. The simulation model is developed, tested and analyzed using the simulation software Arena and SimPy library in Python. The performance measure considered for investigation in the present study is the total patient length of stay which is directly related to patient waiting time and patient satisfaction in the ED. The simulation model was built for the ED of a hospital, and the various bottlenecks in the ED are identified. An integrated metamodel is built in the same Python program using multiple linear regression method. The results are verified and validated. An integrated linear optimization model is developed from the metamodel using PuLP library in Python. The optimal resource allocation is obtained with the objective of reducing the patient length of stay to the desired level within the capacity and budget constraints of the hospital.

It is found that the proposed methodology based on SimPy is comparatively faster than the simulation software Arena. Further, it has the feature of integrating the functions such as simulation, metamodeling and optimization within a single program.

Table 2 Results of MLR model: average waiting time of patients

	Unstandardized Coefficients				95% Confidence Interval for B	
Model	*B*	Std. Error	*t*	Sig.	Lower bound	Upper bound
Constant	54.5621	5.7302	9.5219	0.0000	43.2051	65.9191
Billing counters	0.2438	0.9665	0.2522	0.8013	−1.6718	2.1594
Ultrasound equipment	−0.8748	1.0717	−0.8163	0.4161	−2.9990	1.2493
X-ray, MRI equipment	−13.2730	0.9629	−13.7848	0.0000	−15.1814	−11.3646
Paramedical staff ED doctor	−6.9695	0.9945	−7.0080	0.0000	−8.9406	−4.9984
Stretcher	−0.0519	0.9883	−0.0525	0.9582	−2.0106	1.9068
Surgery ward staff	0.5317	1.0062	0.5284	0.5983	−1.4626	2.5260
Surgeon	0.0706	0.9716	0.0727	0.9422	−1.8551	1.9964
Specialty ward staff	0.1907	0.9650	0.1976	0.8437	−1.7219	2.1033
Specialist	1.7972	1.0053	1.7877	0.0766	−0.1953	3.7896
Orthodoctor and operation theatre staff	−0.1843	1.0013	−0.1840	0.8543	−2.1689	1.8003

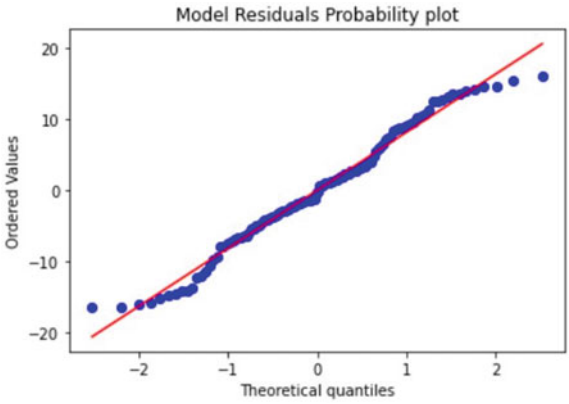

Fig. 4 Normal probability plot of residuals

Thus, it requires minimal interaction from the hospital management staff and aids them in taking quick decisions in times of excess demand. The only input required by the model is the patient inflow data, resource and budget constraints of the hospital. This study can help hospitals in situation like COVID-19 pandemic when the demand grows exponentially. In the future, a time series analysis model can also be integrated

Table 3 Optimal allocation suggested by the model

Name of the resource	Existing quantity	Suggested quantity
Billing counters	3	1
Ultrasound equipment	2	3
X-ray, MRI equipment	2	3
Paramedical staff and	2	2
ED doctor	2	3
Stretcher	1	1
Surgery ward staff	1	1
Surgeon	2	1
Specialty ward staff	1	1
Specialist	1	1
Orthodoctor and operation theatre staff		

with the proposed model to predict the daily patient inflow into the emergency department and to propose the resource allocation for the proceeding days accordingly. Advanced metamodeling techniques like artificial neural networks can be used for more precise results. A significant portion of the patients arriving is related to medico-legal cases which further increases the congestion and waiting time in emergency departments. Applying lean techniques to identify the non-value adding activities at each stage and eliminating them can be performed in the future. The present study is conducted in the emergency department of only one large public hospital in India. The present study can be extended to various other hospitals in different locations.

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